

2 (a) $p = F/A$ M1

use of $m = \rho V$ and use of $V = Ah$ and use of $F = mg$ M1

correct substitution to obtain $p = \rho gh$ A1 [3]

(b) (i) (when h is zero the pressure is not zero due to) pressure from the air/atmosphere B1 [1]

(ii) gradient = ρg or $P - 1.0 \times 10^5 = \rho gh$ C1

e.g. $\rho g = 1.0 \times 10^5 / 0.75 (= 133333)$

$$\rho = 133333 / 9.81$$

$$= 14000 (13592) \text{ kg m}^{-3}$$

A1 [2]

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